Experiment 4: Where the Map Is Wrong

Insight analysis for Chapter 4, Section 4.4

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Purpose of this document. The raw write-up of Experiment 4 (`insights/experiment_4/`) reports that `map/map_tiny.db` places 96.6 % of its occupied cells on real structure and reproduces 89.5 % of the surfaces it observed. This document asks what the remaining 3.4 % and 10.5 % are, and whether the map's quality follows anything a reader would propose: range from the route, how often a surface was observed, the angle it was seen at, the loop-closure correction, the kind of object. It also opens the four unregistered databases that sit beside the scored one. Every number is written by `script/exp4_insights.py` into `output/compare/simulation/exp4-insights/`; the tables below are input from there unchanged. One warehouse and two maps support descriptive patterns, not statistics, and the document says so where it matters. Added 26 October 2026: the same classification applied to the two realistic-warehouse maps of the raw write-up (Insight 6), written by the script's `-realistic` entry into files named `*_realistic.*` beside the tiny ones, which it does not rewrite.

1 Questions and method

- Q1. What are the wrong occupied cells: surfaces drawn too thick, or obstacles where there is nothing?
- Q2. What are the missed surfaces, and does the miss follow range, the number of observing nodes, the viewing angle, or the object?
- Q3. Did the loop closures change the map's accuracy?
- Q4. On an independent route, where does the map block the robot, and why?
- Q5. What are the four unregistered databases, and which recording built the scored map?
- Q6. Do the two realistic-warehouse maps, four times the area and built without loop closure, fail by the same mechanism, and what blocks the independent probe on them?

Raster. The conventions are those of `script/validate_map.py`: the exported occupancy grid is placed in the world by the map pose that `script/fit_map_pose.py` measures from the grid alone, true structure is the world file's obstacle footprints filled plus the wall ring at 0.05 m per cell, true *surfaces* are the one-cell boundary of that, and a cell is correct within 0.25 m. Occupied cells beyond the tolerance are split at 0.50 m: nearer is a surface drawn too thick, further is a phantom. Recall is scored on the surfaces the map observed, as the chapter does.

Per-node observation. RTAB-Map stores, for every graph node, the local occupancy cells it contributed (obstacle, ground and empty, in the node frame). Moving them through the optimised node poses gives, for every world cell, how many nodes observed it and which node observed it first. The incidence angle of a surface is the angle between its outward normal and the line of sight from the nearest graph node: 0° faced head-on, 90° seen edge-on along its own plane.

Second map. `map/map_02.db`, one of the unregistered databases, fits the tiny world as well as the scored map does (Q5) and was built on a different route. It is scored by the same method and used as the

independent drivability probe that the lost `dataset_static_tiny_repeat` reference was meant to provide.

2 Insight 1: the wrong cells are thick surfaces, not phantom obstacles

Of the 3.4 % of occupied cells that miss the tolerance, nine in ten lie within 0.50 m of a real surface, and the rest form 22 clusters of at most ten cells. Table I.1 gives the split and figure I.1 the places.

Table I.1: Occupied map cells by their distance to the nearest true structure (filled shelf footprints and the wall ring of the world file). “Correct” is within the 0.25 m tolerance of the chapter; “thickened” is a cell beyond the tolerance but within 0.50 m of a true surface, i.e. a surface drawn too thick; “phantom” is further than that from any structure. The last columns describe the phantom cells: how many connected clusters they form, the size of the largest, and the median distance of a phantom cell from the recorded route.

Map	occupied cells	correct [%]	thickened [%]	phantom [%]	clusters	largest [cells]	route dist. med. [m]
`map_tiny.db`	11632	96.6	3.0	0.4	22	10	3.18
`map_02.db`	10190	94.3	5.2	0.5	24	16	0.33

The thickened cells sit on the wall ring and on shelf fronts, where the stereo depth of a planar surface spreads over two or three cells. The 43 phantom cells together cover about 0.1 m^2; the largest cluster, ten cells at the north wall near the spawn, is 0.9 m from the route and was observed by one node. None of the phantoms lies in an aisle the recorded route uses, which is why the build route is fully drivable on this map.

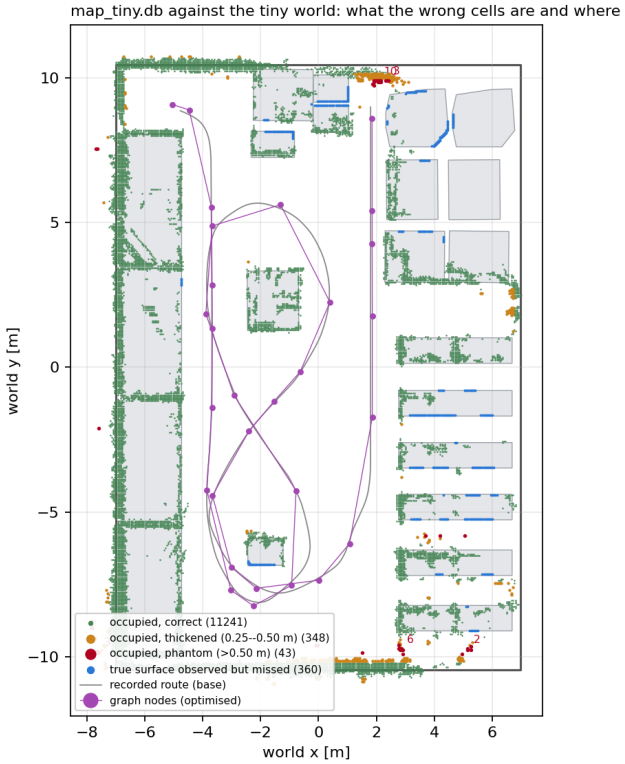


Figure I.1: `map_tiny.db` against the tiny world: correct occupied cells (green), surfaces drawn too thick (orange), phantom cells (red, the largest clusters labelled with their cell count), true surfaces the map observed but did not reproduce (blue), the recorded route and the optimised graph nodes.

3 Insight 2: surfaces are missed when they are seen edge-on and only once, not when they are far

Recall of the observed surfaces falls from 95 % for surfaces faced head-on to 76 % for surfaces seen at more than 60° incidence, and from 97 % for cells observed by three or more nodes to 78 % for cells observed by one; it does not fall with distance from the route within the 6 m the map reaches. Tables I.2 to I.4 and figure I.2 carry the three contrasts.

Table I.2: Recall of the observed surfaces by the incidence angle of the nearest graph node's line of sight on the surface: 0° is a surface faced head-on, 90° one seen edge-on along its own plane (the faces of neighbouring shelves across a narrow gap, looked into from the aisle). Normals come from the world footprints; the viewing node is the nearest optimised graph node.

Incidence angle	`map_tiny`		`map_02`	
[°]	surface cells	recall [%]	surface cells	recall [%]
0–30	1453	95.3	1373	91.8
30–60	1120	92.4	902	90.2
60–90	861	76.1	881	63.1

Table I.3: Precision and recall of the observed surfaces by the number of graph nodes whose stored local grid covers the cell. A map that improves with repeated observation would show both rising with the count.

Observing nodes	`map_tiny` precision		`map_tiny` recall		`map_02` precision		`map_02` recall	
	cells	[%]	cells	[%]	cells	[%]	cells	[%]
1	2529	91.2	834	78.2	2821	96.1	1166	74.1
2	3631	96.8	854	90.5	3018	92.2	773	93.1
3–4	4642	99.4	1246	97.4	2452	92.0	421	98.8
5–7	661	99.7	238	97.9	37	100.0	21	81.0
≥ 8	1	100.0	0	—	0	—	0	—

Table I.4: Precision and recall of the observed surfaces by distance from the recorded route (`dataset_map_tiny`, the route that built `map_tiny.db`; for `map_02.db` the same route is only a proxy for its own, unknown route). Stereo depth error grows with range, so a range effect would show as falling precision and recall with distance.

Distance from route	`map_tiny` precision		`map_tiny` recall		`map_02` precision		`map_02` recall	
[m]	cells	[%]	cells	[%]	cells	[%]	cells	[%]
0–1	819	97.3	436	89.0	723	90.0	398	85.7
1–2	3211	96.4	932	88.7	2824	97.1	872	85.7
2–3	3389	97.4	836	86.0	3023	91.9	778	77.6
3–4	3183	97.6	862	94.3	2869	96.2	816	86.9
4–6	1015	91.1	364	88.7	730	91.0	281	77.9
6–8	15	100.0	4	100.0	21	76.2	11	100.0
≥ 8	0	—	0	—	0	—	0	—

The two mechanisms are the same surfaces seen from the same place. The missed cells in figure I.1 are the faces of neighbouring shelves across the narrow gaps on the east side, which the route looks into from the aisle but never enters: each face is seen once, from a node in the aisle, at grazing incidence. Precision tells the same story from the other side: cells observed by one node are 91 % correct, cells observed by three or more 99 %. Stereo range error, the mechanism a reader would propose first, does not show: precision stays at 96–98 % out to 4 m and recall has no trend with distance.

Table I.5 lists the same effect by object. The objects with the lowest recall were each observed by a single node, while the one large shelf observed by four nodes is reproduced at 99 %. Which objects these are follows from the route, not from the objects: the figure-eight and the one straight along the east wall decide what is faced and what is grazed.

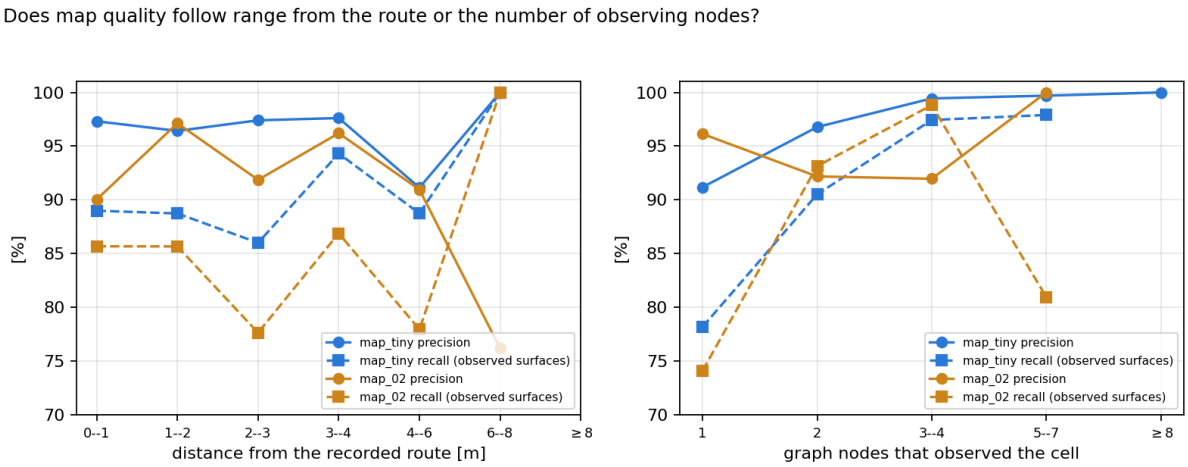


Figure I.2: Precision of the occupied cells and recall of the observed surfaces against distance from the recorded route (left) and against the number of graph nodes that observed the cell (right), for both tiny-warehouse maps.

Table I.5: Recall of `map_tiny.db` on the observed surfaces of each world object, worst first: surface cells the map observed, their share of the object's surface, the share of them that the map reproduces within tolerance, the median distance of those cells from the recorded route, and the median number of graph nodes that observed them.

Object	observed [cells]	of surface [%]	recall [%]	route dist. med. [m]	nodes med.
ClutteringA_01_016	10	6	0.0	2.9	1
ShelfD_01_001	103	55	48.5	1.9	1
ClutteringA_01_017	107	66	52.3	1.1	1
Bucket_01_020	68	100	52.9	1.5	1
ShelfE_01_002	159	85	69.2	2.8	1
ShelfD_01_002	128	68	69.5	2.3	1
Bucket_01_022	82	100	72.0	1.4	2
TrashCanC_01_002	90	100	72.2	1.8	3
Bucket_01_021	84	100	75.0	0.9	2
ClutteringC_01_027	114	75	84.2	1.1	3
ShelfD_01_003	148	80	91.2	3.2	2
ClutteringC_01_030	82	55	91.5	0.6	3
ShelfE_01_003	156	83	94.2	4.1	1
ClutteringC_01_032	138	100	96.4	2.3	2
ShelfF_01_001	802	100	99.4	2.4	4
ShelfE_01_001	169	90	100.0	2.7	2
ClutteringC_01_028	67	44	100.0	3.6	1
ClutteringC_01_031	150	100	100.0	1.2	4
wall ring	777	56	100.0	3.2	2

4 Insight 3: the loop closures corrected the graph, and the map did not need it

The five closure links corrected the odometry by at most 0.26 m in the plane and 0.52 m in height, and the occupied cells first observed after the closing revisit are as accurate as those observed before it. Table I.6 and figure I.3 carry the numbers.

Table I.6: Loop closures of `map_tiny.db` and the correction the graph optimisation applied: planar and vertical distance between each node's odometry pose and its optimised pose, by node range, and the precision of the occupied cells first observed before and after the revisit that closed the loop (nodes 1–18 against 19–29).

Quantity	nodes 1–18	nodes 19–29	all
planar correction, median [m]	0.01	0.16	0.09
planar correction, max [m]	0.12	0.26	0.26
vertical correction, max [m]	0.18	0.52	0.52
occupied cells first seen	10098	1366	11632
precision of those cells [%]	96.9	95.6	96.6
closure links (from → to): 9→22, 10→23, 22→9, 23→10, 30→29			

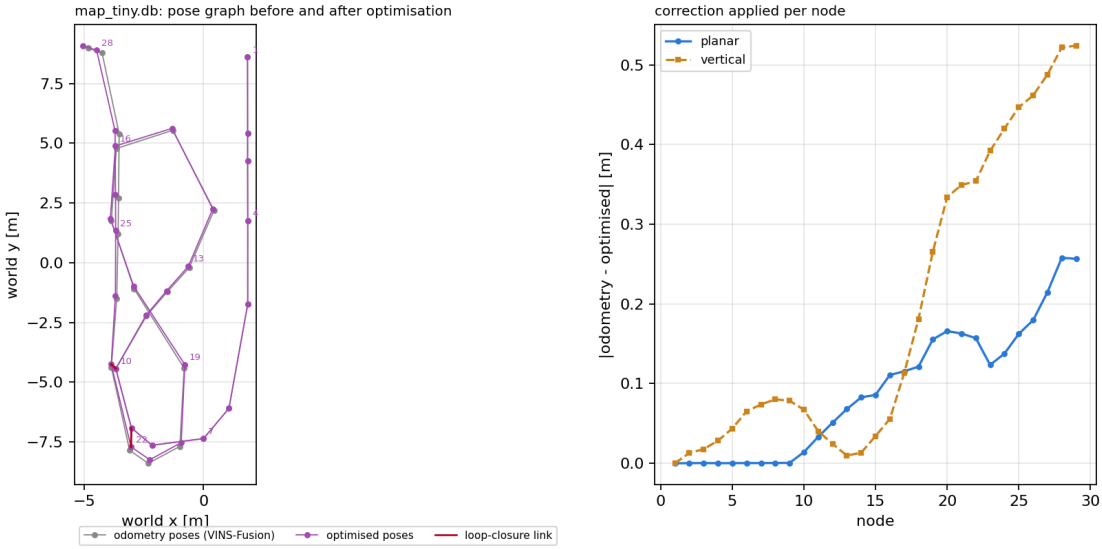


Figure I.3: `map_tiny.db`: odometry poses and optimised poses of the 29 nodes in the optimised graph, with the loop-closure links (left), and the planar and vertical correction each node received (right).

The correction is real but small, and it is mostly vertical: VINS-Fusion's odometry climbed 0.5 m over the 70 m route, and the closures at nodes 22 and 23, which revisit nodes 9 and 10, pulled the second half of the graph back onto the floor. In the plane the correction stays under the 0.25 m tolerance for every node, so the cells placed from corrected and from uncorrected poses score alike. On a route of this length the map's accuracy was set by the odometry, with the closures as insurance that was not drawn on.

5 Insight 4: on an independent route the second map blocks the robot at its phantom cells

Under the true footprint along the recorded route, `map_02.db` blocks 7.6 % of the poses and leaves 5.5 % unmapped, and every blocked stretch coincides with one of its phantom clusters. Tables I.7 and I.8 and figure I.4 carry the evidence.

Table I.7: Drivability of both tiny-warehouse maps under the nav2 footprint stamped along the recorded route of `dataset_map_tiny` (3032 poses). For `map_tiny.db` this is the route that built the map; for `map_02.db` it is an independent route through the same world, which the lost `dataset_static_tiny_repeat` reference was meant to provide.

Map	route	false obstacle [% of poses]	unmapped [%]	clearance med. [m]	clearance p05 [m]
`map_tiny.db`	built the map	0.0	0.0	1.01	0.46
`map_02.db`	independent	7.6	5.5	0.98	0.25

Table I.8: Contiguous stretches of the recorded `dataset_map_tiny` route on which the true footprint overlaps an occupied cell of the map: pose range, number of poses, location, and the smallest clearance from the route centre to a mapped obstacle inside the stretch.

Map	poses	count	x [m]	y [m]	min. clearance [m]
`map_tiny.db`	—	0	—	—	—
`map_02.db`	2356–2421	66	-3.7	-5.9	0.10
`map_02.db`	125–164	40	1.8	8.2	0.00
`map_02.db`	922–958	37	-3.6	-5.8	0.00
`map_02.db`	1485–1516	32	-2.4	5.6	0.11
`map_02.db`	960–986	27	-3.8	-5.2	0.20
`map_02.db`	1518–1539	22	-2.8	5.4	0.05
`map_02.db`	916–918	3	-3.4	-6.2	0.40
`map_02.db`	1406–1407	2	-0.7	5.1	0.54

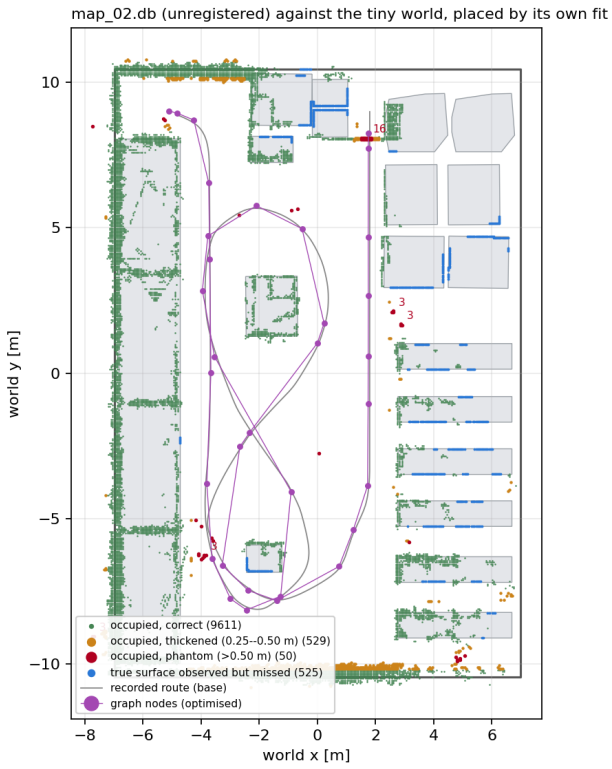


Figure I.4: `map_02.db`, unregistered, against the tiny world, placed by its own fit; same conventions as figure I.1. The recorded route drawn is that of `dataset_map_tiny`, which this map did not see.

This is the measurement the raw write-up could not make. The build route is drivable on the map it built by construction, since the camera cleared the cells the robot passed through; an independent route meets the cells the build route only looked at. On the second map the three blocked stretches sit on its phantom clusters of 3, 16 and 3 cells, in the figure-eight and at the start of the east straight; the route centre comes within 0.0 m to 0.2 m of them. Phantoms that cover 0.1 m^2 of a 290 m^2 floor are negligible for geometry and decisive for drivability when they fall in an aisle.

[Placement of the second map] `map_02.db` is placed by its own grid fit (agreement 0.385, margin 54 % over the best rival yaw), not by a known spawn. A placement error of a few centimetres moves the thickened wall cells but not the phantom clusters, so the blocked stretches stand; the per-cell precision of this map carries that uncertainty.

6 Insight 5: the scored map was built from the recorded tiny bag, and the large-warehouse attempt left a database

The nodes of `map_tiny.db` coincide with the ground truth of `dataset_map_tiny` at the same stamps to a median 0.05 m, so the map was built from that recording and the write-up's self-check is a true self-check; and `map_03.db` is a 76 m run whose poses climb to 9.8 m above the floor, the large-warehouse attempt that the chapter reports as having left no artefact. Table I.9 describes all five databases.

Table I.9: The five RTAB-Map databases under `map/`: size, graph nodes, simulation-time span of the nodes, odometry path length and planar extent, vertical range of the node poses, loop closures, the placement fit against the tiny world (agreement score and margin over the best rival yaw, from `script/fit_map_pose.py`), and what the database is identified as. Only `map_tiny.db` is registered in the project records.

Database	MB	nodes	stamps [s]	path [m]	extent [m]	z [m]	closures	fit (agr., margin)	identified as
`map_tiny.db`	67.4	30	31–90	69.5	17×7	-0.0..0.6	5	0.387, +61 %	the scored tiny map (registered); built on the `dataset_map_tiny` route
`map_01.db`	0.8	2	59–61	0.0	0×0	-0.1..1.7	0	—	aborted start (2 nodes)
`map_02.db`	55.3	38	30–91	68.8	7×17	-0.3..-0.1	7	0.385, +54 %	a second tiny-warehouse map on a different route (21 Sep)
`map_02_vins.db`	9.3	2	34–35	0.7	1×0	0.1..0.2	0	—	aborted start (2 nodes)
`map_03.db`	29.2	18	35–64	75.8	5×37	-0.3..9.8	0	0.19, +7 % (large worlds; weak)	a 37 m straight run that left the floor: the lost large-warehouse attempt, unplaceable

`map_02.db` fits the tiny world as well as the scored map does and its nodes do not follow the recorded route (median 4.1 m from the ground truth at the same stamps): it is a second map of the same world on a different route, made on 21 September, and the only candidate for the lost `dataset_static_tiny_repeat` session. `map_03.db` cannot be placed in any large world (agreement 0.19, margin 7 %); its 18 nodes run 37 m along one aisle and leave the floor, which is tracking failure, not mapping failure. `map_01.db` and `map_02_vins.db` are two-node starts.

[Provenance] The identifications of `map_02.db` and `map_03.db` rest on their contents alone; no record names them. The raw write-up's statement that the scored map's node stamps match no recorded dataset is contradicted by the stamp-and-pose match above and should be withdrawn.

7 Insight 6: on the realistic maps the shelves are thickened, the walls are displaced, and the cells that block the probe were laid during initialisation

Of the occupied cells of the two realistic maps, 13.4 and 9.7 % are surfaces drawn too thick and 5.2 and 5.3 % are phantoms, thirteen times the tiny map's phantom share; every probe pose that the maps block lies within 0.6 m of a phantom cell, and 96 and 63 % of those blocking cells were first observed by one of the first five graph nodes. Table I.10 gives the split, figures I.5 and I.6 the places, and tables I.14 and I.15 the blocked stretches.

Table I.10: Occupied map cells by their distance to the nearest true structure (filled shelf footprints and the wall ring of the world file). "Correct" is within the 0.25 m tolerance of the chapter; "thickened" is a cell beyond the tolerance but within 0.50 m of a true surface, i.e. a surface drawn too thick; "phantom" is further than that from any structure. The last columns describe the phantom cells: how many connected clusters they form, the size of the largest, and the median distance of a phantom cell from the independent probe route.

Map	occupied cells	correct [%]	thickened [%]	phantom [%]	clusters	largest [cells]	route dist. med. [m]
`map_realistic.db`	26711	81.3	13.4	5.2	413	108	4.89
`map_realistic_01.db`	25395	85.0	9.7	5.3	385	157	10.10

The thickening is the tiny map's mechanism at the shelves, and the phantoms are a different one at the walls. In both figures the shelf rows the build route passed are green with an orange rim, surfaces spread over two or three cells as on the tiny map; the phantoms form a continuous red band the full length of the east wall about 0.6 m inside its true line, and a second band along the south wall outside it, in both maps alike. The largest clusters, 108 and 70 cells on the U map and 157 and 117 on the I map, all sit on that east band, 6 to 11 m from the probe route and first observed at median node 32 and 36. The walls are the structure furthest from any build node; what moved them is not identified here.

Table I.11: Precision and recall of the observed surfaces by the number of graph nodes whose stored local grid covers the cell, for the two realistic-warehouse maps (U is `map_realistic`, I is `map_realistic_01`). The tiny-map contrast was 91 to 99 percent precision and 78 to 97 percent recall from one node to three or more.

Observing nodes	`U` precision cells	[%]	`U` recall cells	[%]	`I` precision cells	[%]	`I` recall cells	[%]
1	4891	65.5	1884	71.8	4068	83.2	1788	75.8
2	6048	75.3	1828	81.8	6236	80.3	2153	79.8
3–4	7958	82.0	2967	86.1	8006	82.9	3017	82.4
5–7	5261	98.9	2708	90.1	4992	93.2	2296	85.3
≥ 8	1274	100.0	840	91.0	1315	99.9	732	87.7

Table I.12: Recall of the observed surfaces by the incidence angle of the nearest graph node's line of sight on the surface: 0° is a surface faced head-on, 90° one seen edge-on along its own plane (the faces of neighbouring shelves across a narrow gap, looked into from the aisle). Normals come from the world footprints; the viewing node is the nearest optimised graph node.

Incidence angle [°]	`map_realistic` surface cells	`map_realistic` recall [%]	`map_realistic_01` surface cells	`map_realistic_01` recall [%]
0–30	6042	85.4	5445	82.6
30–60	2413	84.1	2535	82.1
60–90	2454	74.7	2477	78.4

The view-count mechanism of Insight 2 holds and is larger. Cells observed by one node are 66 and 83 % correct, cells observed by five or more 99 and 100 %, against 91 and 99 % on the tiny map; recall of

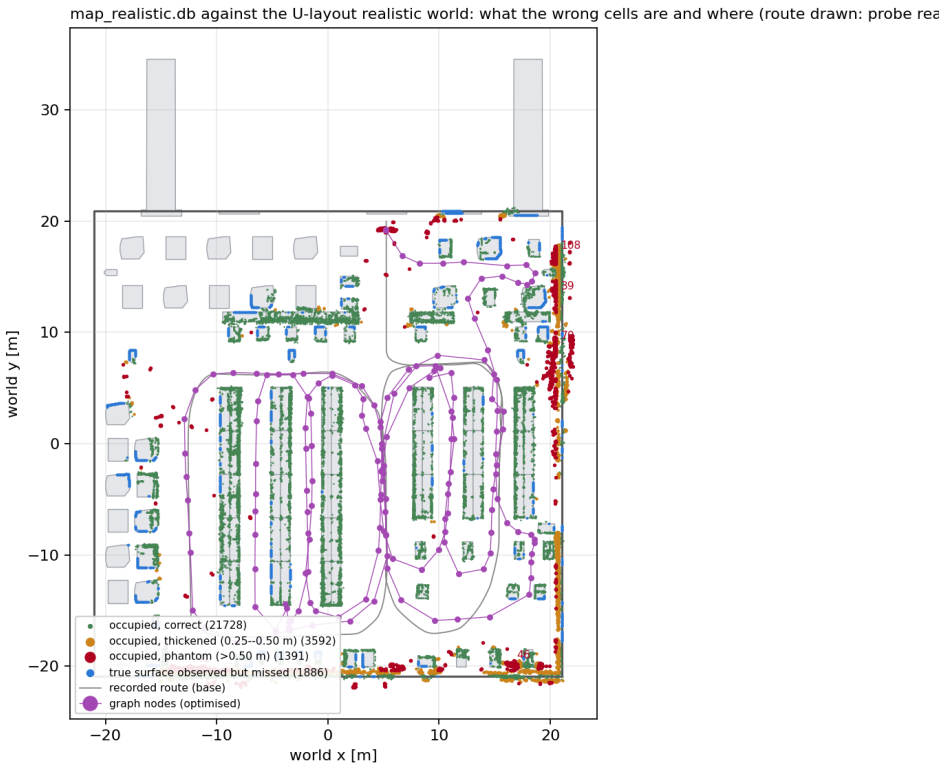


Figure I.5: `map_realistic.db` against the U-layout realistic world, placed by its fit; conventions as figure I.1. The route drawn is the independent probe `realistic_000`, which did not build the map; the graph nodes are the build route. Phantom clusters are labelled with their cell count.

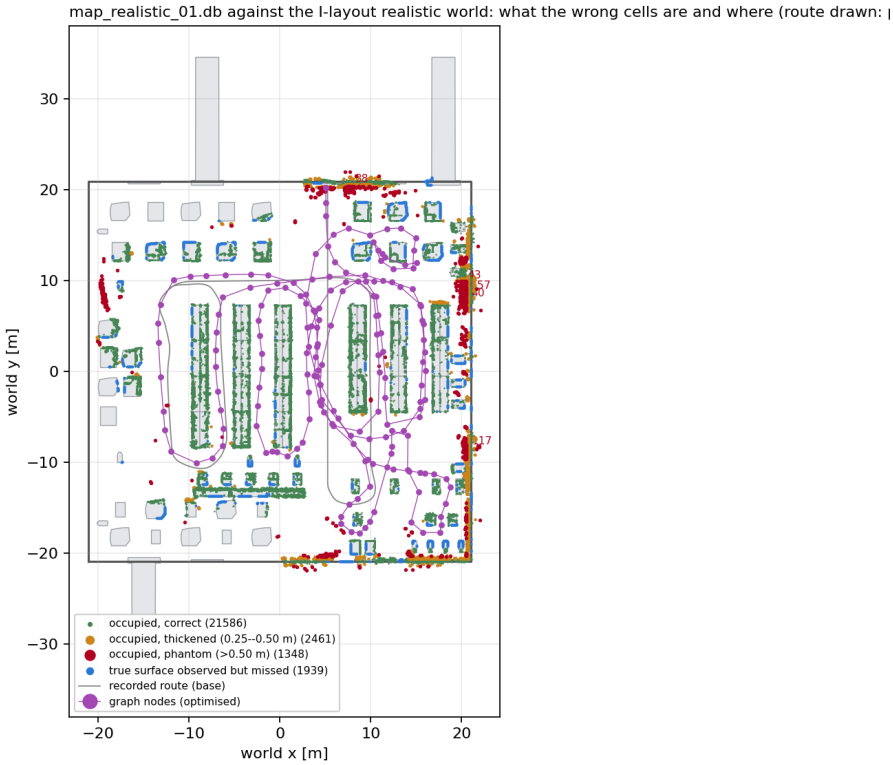


Figure I.6: `map_realistic_01.db` against the I-layout realistic world, placed by its fit; the route drawn is the probe `realistic_001`.

Table I.13: Precision and recall of the observed surfaces by distance from the independent probe route (`realistic_000` for U, `map_realistic`; `realistic_001` for I, `map_realistic_01`). The probe is not the route that built either map, so this is only a proxy for range from the mapping camera.

Distance from route [m]	U precision cells	U precision [%]	U recall cells	U recall [%]	I precision cells	I precision [%]	I recall cells	I recall [%]
0–1	110	6.4	1	100.0	1281	91.7	415	95.2
1–2	2343	99.0	1142	82.4	2581	96.6	907	95.0
2–3	4387	97.7	2248	88.5	5172	96.3	1875	90.7
3–4	6575	80.5	2130	87.1	2655	96.5	1200	80.4
4–6	8608	81.5	3369	82.4	4475	95.8	2188	78.0
6–8	2202	64.9	1182	68.9	3073	84.5	1497	80.4
≥ 8	2486	55.3	837	77.1	6158	56.7	2375	70.9

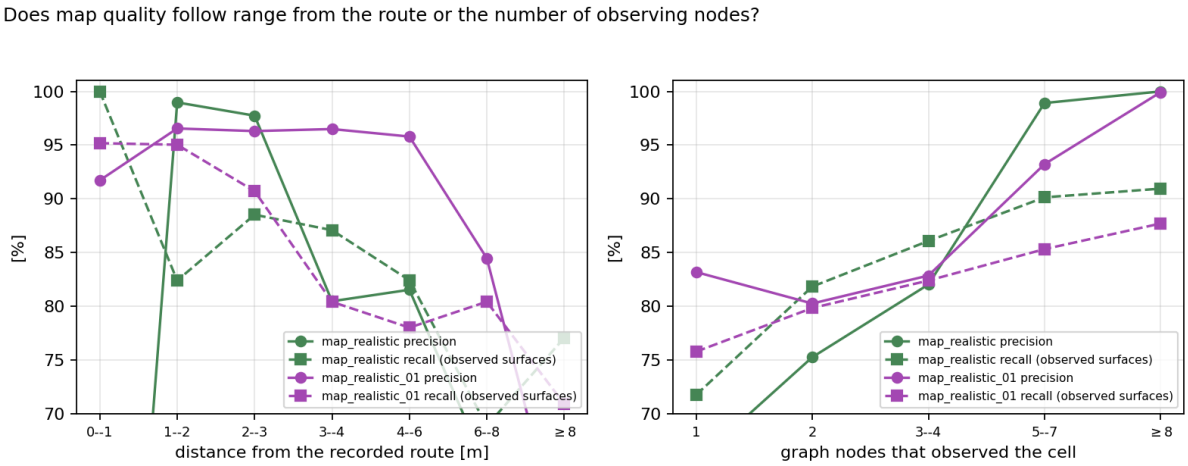


Figure I.7: Precision of the occupied cells and recall of the observed surfaces against distance from the independent probe route (left, a proxy only: the probe did not build either map) and against the number of graph nodes that observed the cell (right), for the two realistic-warehouse maps.

the observed surfaces rises from 72 and 76 % at one node to 91 and 88 % at eight or more. The incidence effect also holds, 85 and 83 % head-on against 75 and 78 % grazing, but the head-on figure is ten points below the tiny map's 95 %, so viewing geometry explains the spread between surfaces and not the level they all sit at.

Table I.14: Drivability of the two realistic-warehouse maps under the nav2 footprint stamped along the static-twin recording of each layout, which did not build the map (`realistic_000`, 6278 poses, for the U map; `realistic_001`, 4847 poses, for the I map). Same method as the raw write-up, which it reproduces.

Map	route	false obstacle [% of poses]	unmapped [%]	clearance med. [m]	clearance p05 [m]
`map_realistic.db`	independent	2.1	2.4	1.95	0.61
`map_realistic_01.db`	independent	4.4	0.7	2.16	0.36

Table I.15: Contiguous stretches of the independent probe route of each layout on which the true footprint overlaps an occupied cell of the map: pose range, number of poses, location, and the smallest clearance from the route centre to a mapped obstacle inside the stretch.

Map	poses	count	x [m]	y [m]	min. clearance [m]
`map_realistic.db`	108–184	77	5.2	19.5	0.00
`map_realistic.db`	3332–3360	29	-12.5	1.7	0.21
`map_realistic.db`	284–311	28	5.2	15.3	0.00
`map_realistic_01.db`	0–149	150	5.2	20.0	0.20
`map_realistic_01.db`	1836–1868	33	10.0	-3.2	0.00
`map_realistic_01.db`	3431–3459	29	-12.2	-3.6	0.15
`map_realistic_01.db`	3729–3729	1	-9.2	-10.6	0.40

The probes are blocked by the phantoms, not by the thickened shelving, and the blocking phantoms are the oldest cells in each map. Of 134 and 213 blocked poses, 134 and 212 lie within 0.6 m of a phantom cell and none within that distance of a thickened cell alone. The largest stretch on each map is the first metres of the probe at the spawn, 77 and 150 poses; the 77 and 41 phantom cells under the blocked stretches have a median first-observing node of 1, and 96 and 63 % of them were first seen by node 5 or earlier, against 6 and 3 % of all phantom cells. The raw write-up records that node 1 of each database is not at identity and that its height settles over the first nodes, the inertial initialisation transient; the cells laid during that transient are the ones under the robot's starting position, which is where every probe begins. The remaining stretches, at the left shelf column on both maps and in the one-aisle gap at $x \approx 10$ on the I map, sit on small phantom clusters laid mid-run.

[Two rasters, one disagreement] The chapter's `validate_map.py` reports 3.8 % blocked poses on `map_realistic`; this document's raster, whose origin is the wall ring rather than the union of map, route and wall, reports 2.1 %. The difference is the spawn stretch alone: the chapter counts probe poses 0–183, this raster 108–184, and the route centre there has 0.00 m clearance, so the count flips on a sub-cell offset. The other two stretches, `map_realistic_01` in full (4.4 against 4.4 %), and every clearance and unmapped figure agree to 0.01 m. The chapter's figure stands as the registered one; the spawn-stretch count carries about 1.7 % of raster sensitivity.

[Wall displacement, untested] The east band is inside the wall and the south band outside it in both maps, which a uniform scale error would not produce; drift accumulated along the closure-free build route is the candidate, but no test separates it from a range-dependent stereo bias at the 3 to 5 m the walls were seen from. The build-route ground truth on the removable drive would decide it.

8 What these data cannot answer

- **Whether the patterns generalise.** Two maps of one small world; the contrasts are descriptive. The view-count and incidence effects are consistent between the two maps and with how stereo mapping works, which is the reason to trust them, not a test.
- **The old large-warehouse attempt.** `map_03.db` exists but cannot be scored: no placement fits, and the pose graph left the floor. The two realistic-warehouse maps of Insight 6 now stand as the large result; whether their patterns hold under loop closure is untested, because they were built without it.
- **Whether the second map's phantoms are typical.** Their cause is not identified; they appear where the second map's own route started and turned, which suggests transient depth at initialisation, untested.
- **Distance beyond 6 m.** The map reaches no further, so the absence of a range effect is bounded to that range.

9 Summary

Table I.16: Questions, answers and the strength of the evidence.

	Answer	Evidence	Confidence
Q1 wrong cells	Thick surfaces, not phantoms: 348 thickened against 43 phantom cells in 22 small clusters	table I.1, figure I.1	high
Q2 missed surfaces	Seen edge-on and once: recall 95 % head-on to 76 % grazing, 97 % at three nodes to 78 % at one; no range effect within 6 m	tables I.2 to I.4	medium, two maps
Q3 loop closures	Corrected 0.26 m planar and 0.52 m vertical; map accuracy unchanged across the revisit	table I.6	medium
Q4 independent route	Second map blocks 7.6 % of poses, all at its phantom clusters	tables I.7 and I.8	medium, placement by fit
Q5 provenance	Scored map built from `dataset_map_tiny`; `map_02` a second tiny map; `map_03` the large attempt, unscorable	table I.9	high / medium
Q6 realistic maps	Shelves thickened as on the tiny map; walls displaced about 0.5 m as phantom bands; probes blocked only at phantoms, 96 % / 63 % of them laid by node 5	tables I.10 and I.15	medium, one map per layout

Consequences for Chapter 4. Report the map's error as surface thickness plus a handful of phantom cells, and the recall gap as a viewing-geometry effect of the route, not a sensor-range effect. Quote the independent-route drivability from `map_02.db` on the recorded route, with its provenance stated,

in place of the lost reference. Correct the statement that the scored map matches no recorded dataset. Report `map_03.db` as the retained artefact of the large-warehouse attempt and its failure as tracking loss, with the database available for a future placement attempt. For the realistic maps, attribute the false obstacles to phantom cells laid during initialisation and at the walls, not to thickened shelving, and state that discarding the cells of the first few nodes, or starting the map after the inertial transient has settled, is the one correction the data predict would remove the spawn stretch, 57 and 70 % of the blocked poses.